

44% compared to GPUs while delivering competitive PPW. Instead of using expensive high-bandwidth memory, the MTIA uses large static random-access memory (SRAM) alongside low-power double data rate memory. Instead of focusing only on architectural design, the team has co-designed and optimised their ML models to work effectively with the chip.

Microsoft Maia: Microsoft's Maia 100 is a custom AI accelerator designed to handle the massive AI workloads across Azure. It is built with 105 billion 5-nanometre transistors, high-bandwidth mesh interconnect, large on-chip SRAM to reduce external memory trips, and liquid cooling for dense, energy-efficient deployments. It connects via Azure's custom network fabric with integrated smart network interface unit data processing units to offload networking and security. The co-designed software stack supports popular frameworks and graph-level optimisations to improve utilisation. Microsoft emphasises that the end-to-end co-design of the chip, system, data centre, and software helped bring down TCO and improve performance, power-efficiency, and sustainability for foundation models and enterprise AI services across Azure.

Intel: Intel became active in the AI accelerator market with the acquisition of the Gaudi series from Habana Labs. Gaudi 3 is a 5nm AI accelerator family purpose-built to train and serve GenAI and LLMs at scale. The architecture uses matrix multiplication engines, tensor cores, 128GB high-bandwidth memory and on-die SRAM to deliver GPU-class throughput. Integrated 200GB Ethernet lets clusters scale to hundreds of nodes without external switches. The AI accelerators integrate with the Intel oneAPI ecosystem, allowing developers to easily develop and deploy AI models using open source software and frameworks such as PyTorch, TensorFlow, DeepSpeed, and Hugging Face.

AMD Instinct and Ryzen: AMD has been aggressively pushing its AI offerings, such as the Instinct MI300 and MI400 accelerators for large-scale data centres and AI training, and the Ryzen AI processors for client devices, with an integrated neural processing unit for on-device AI tasks. Its ROCm (Radeon Open Compute) software platform is also improving and gaining traction, focusing on open source support.

Tesla AI5 and AI6: Tesla is also developing its own AI chips, targeting lower costs and better PPW. The chips are in the design review stage, and Tesla will be working with TSMC and Samsung for their manufacture.

OpenAI: OpenAI is working with Broadcom to produce its own proprietary AI chip to secure guaranteed compute capacity without depending on Nvidia's supply of GPUs. The first chips are expected to roll out in 2026 and will first cater to a massive \$10 billion internal order.

Huawei's Ascend series: China has been encouraging local companies like Huawei and Alibaba to develop indigenous AI chips to build a domestic AI infrastructure that is less dependent on Nvidia's high-end chips. Huawei's Ascend series of chips, namely Ascend 910B and 910C, are in production and being used to power various AI applications within China. The company is currently facing manufacturing delays due to issues in sourcing components, such as high-bandwidth memory (HBM), from the US. In the meanwhile, they are continuing to develop and test the next generation of Ascend 910D chips. Alibaba has also developed its own AI chip, which is currently under testing.

Made-in-India GPUs, coming soon: According to a recent announcement by Ashwini Vaishnaw, Union Minister for Electronics and Information Technology, India is working towards developing its own GPU within the next three to five years.

It also hopes to have its own indigenous foundational AI model ready within a year. In the meantime, the Indian startup ecosystem has introduced several innovations, including running AI on standard CPUs and developing edge AI chips.

Other innovative AI chips

Apart from the hyperscalers, some startups have developed interesting AI chips from the ground up, challenging the dominance of general-purpose GPUs. These may not pose a big threat to Nvidia as yet, but the concept has taken root quite strongly.

Groq: Californian startup Groq has built hardware designed from the ground up for superfast AI inference for LLMs, and other complex AI tasks like natural language processing (NLP), image analysis, and complex computations at scale. Groq's offering centres around its unique language processing unit and the GroqChip, which together deliver higher inference speeds and lower latency than general-purpose GPUs. Developers can access Groq's technology through its free-to-use GroqCloud API, which is compatible with popular AI development frameworks.

Cerebras: An American startup that also has a presence in Bengaluru, Cerebras' Wafer-Scale Engine is a large chip built specifically for ultra-fast AI. It costs much less than GPU clouds, but delivers 30x faster inference, according to the company. While the company initially started with large chips for AI training, they recently shifted their focus to AI inference. Cerebras Inference, which they launched in 2024, delivers 1800 tokens per second for Llama 3.1 8B, and 450 tokens per second for Llama 3.1 70B. In August this year they announced that they will help power OpenAI's open model gpt-oss-120B. Beyond inference, clients can also use the Cerebras platform to optimise AI models for their specific use-cases by fine-tuning or pre-training